

4. Current Electricity and Magnetism

1. Write proper words from the following group of words in the blanks and rewrite the completed sentences:

(magnetism, 4.5V, 3.0V, gravitational attraction, potential difference, potential, higher, lower, 0V)

(a) Water in the waterfall flows from a higher level to the lower level because of

Answer:

Water in the waterfall flows from a higher level to the lower level because of gravitational attraction.

(b) In an electric circuit, electrons flow from a point of potential to the point of potential.

Answer:

In an electric circuit, electrons flow from a point of lower potential to the point of higher potential.

(c) The difference between the electrostatic potential of the positive end and the negative end of an electric cell is the of the cell.

Answer:

The difference between the electrostatic potential of the positive end and the negative end of an electric cell is the potential difference of the cell.

(d) Three electric cells of potential difference 1.5 V each have been connected as a battery. The potential difference of the battery will be V.

Answer:

Three electric cells of potential difference 1.5 V each have been connected as a battery. The potential difference of the battery will be 4.5 V.

(e) An electric current flowing in a wire creates around the wire.

Answer:

An electric current flowing in a wire creates magnetism around the wire.

2. A battery is to be formed by joining 3 dry cells with connecting wires. Show how you will connect the wires by drawing a diagram.

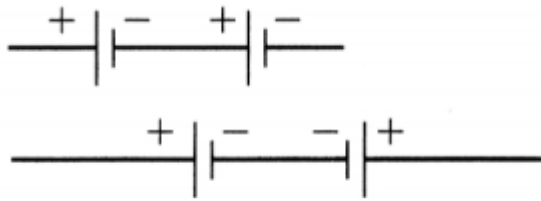
Answer:



3. In an electric circuit, a battery and a bulb have been connected and the battery consists of two cells of equal potential difference. If the bulb is not glowing, then which tests will you perform in order to find out the reason for the bulb not glowing?

Answer:

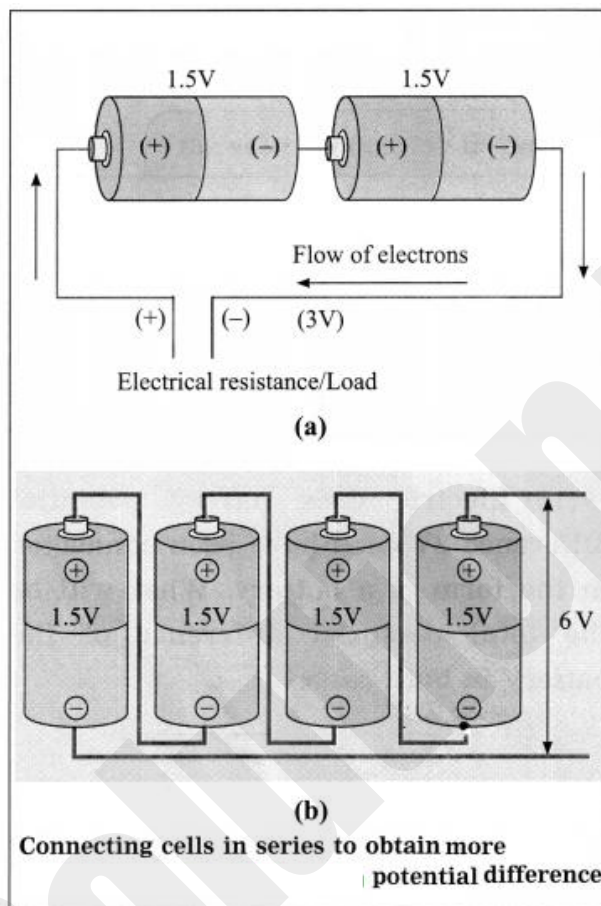
If you can see the filament of the bulb, check whether it is intact or broken. Check whether the cells are connected in a proper manner:



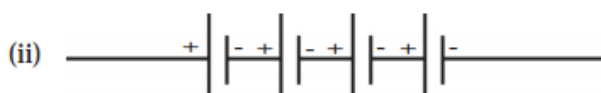
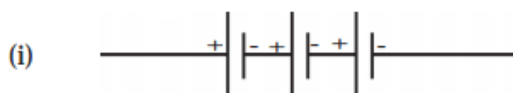
Or in a wrong way:

If they are connected in a wrong way as shown above, the total potential difference will be $2V + (-2V) = \text{zero}$.

Figures for reference:



4. Electric cells having potential difference 2V each have been connected in the form of a battery. What will be the total potential difference of the battery in both cases?



Answer:

(i) 6V

(ii) 8V.

[Note: In (i), three cells are connected in series.

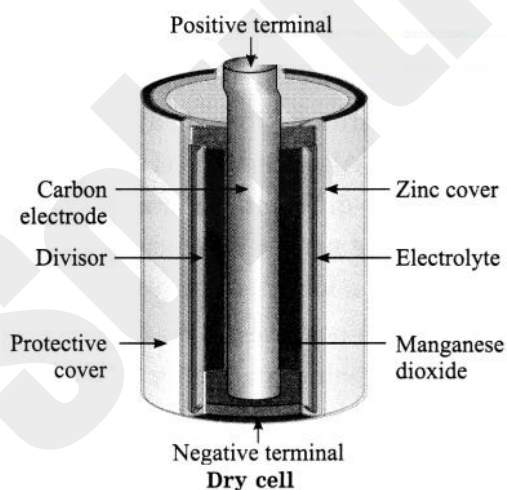
$\therefore$ Total potential difference = $2V + 2V + 2V = 6V$.

In (ii), four cells are connected in series.

$\therefore$ Total potential difference = $2V + 2V + 2V + 2V = 8V$]

5. Describe the construction, working and usefulness of a dry cell, with the help of a diagram.

Answer:



Dry cell:

Take a lead dry cell and remove its outer coating. Inside you will find a whitish, metal layer. This is the zinc (Zn) metal layer. This is the negative terminal of the cell. Now, carefully break open this layer. There is another layer inside. An electrolyte is filled between these two layers.

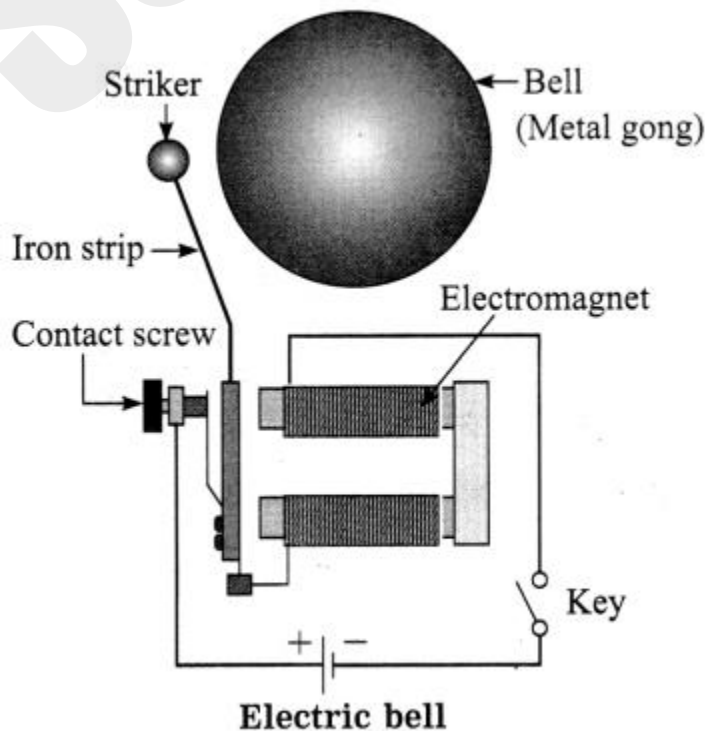
The electrolyte contains negatively charged and positively charged ions. These are the carriers of electricity. The electrolyte is a wet pulp of zinc chloride (ZnCl_2) and ammonium chloride (NH_4Cl). There is a graphite rod at the centre of the cell. This is positive terminal of the cell. A paste of manganese dioxide (MnO_2) is filled outside the rod. Because of the chemical reactions of all these chemicals, electrical charge is produced on the two terminals (graphite rod and zinc layer) and an electric current flows in the circuit.

Due to the wet pulp used in this cell, the chemical reaction proceeds very slowly. Hence a large electric current cannot be obtained from this. Compared to the electric cells using liquids, the shelf life of dry cells is longer. Dry cells are very convenient to use as these can be held in any direction with respect to ground and can be used in mobile instruments such as radio sets, wall clocks and torches.

6. Describe the construction and working of an electric bell with the help of a diagram.

Answer:

Figure shows the construction of an electric bell and also the electric circuit. It consists of an electromagnet, contact screw, iron strip, metal striker and metal gong. A coil of copper wire wound around an iron piece works as an electromagnet and an iron strip along with a striker is fitted near it.



The circuit is closed by key. The current flows in the circuit when the screw is in contact with the iron strip. The current – carrying coil becomes a magnet and attracts the iron strip towards it. As a result, the metal striker hits the metal gong producing sound. At the same time, the screw loses the contact with the strip. The circuit is now incomplete.

Thus, no current flows in the circuit.,and the electromagnet loses its magnetism and the electric iron strip returns to its initial condition, making contact with the screw. As the circuit is now completed, the electromagnet attracts the iron strip and the striker strikes the gong producing sound.

The action repeats & bell continues to ring till the circuit is broken by opening the key in the circuit.

The working of the electric bell is hence based on the magnetic effect of electric current.